Week 3 – day 2 – Assignment

With vibe coding implement the below

1)Connect to jira via api

2)Fetch the user storie from jira to generate testcases

Front End and Backend implemented

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EXPLORER

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- backend
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OUTLINE
TIMELINE

.env | Chat

Describe what to build

Agent | Claude Haiku 4.5

Local | Default Approvals

CHAT

JIRA INTEGRATION IMPLEMENTATION PLAN

🎉 JIRA INTEGRATION COMPLETE! ✅

I've successfully implemented end-to-end Jira integration following your requirements. Here's what was delivered:

📋 Phase 1: Backend Implementation ✅

Created Files:

1. jiraClient.ts
   - HTTP client for Jira REST API using Axios
   - Methods: `getProjects()`, `getUserStories()`, `getUserStory()`, `testConnection()`
   - Error handling with custom `JiraClientError` class
   - Basic Auth support (email + API token)
2. jiraConfig.ts
   - Configuration manager with singleton pattern
   - Methods: loadFromEnvironment() , `setConfig()`, `getClient()`, `validateConnection()`
   - Secure credential management
3. jira.ts
   - 5 REST API endpoints:

Todos (12/12)

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2. jiraConfig.ts
   - Configuration manager with singleton pattern
   - Methods: loadFromEnvironment() , `setConfig()`, `getClient()`, `validateConnection()`
   - Secure credential management
3. jira.ts
   - 5 REST API endpoints:
     - POST /api/jira/config - Configure credentials
     - GET /api/jira/config/status - Check status
     - GET /api/jira/projects - Fetch all projects
     - `GET /api/jira/projects/:projectKey/user-stories` - Fetch user stories
     - GET /api/jira/issues/:issueKey - Fetch specific issue

Updated Files:

Todos (12/12)

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JIRA INTEGRATION IMPLEMENTATION PLAN

- GET /api/jira/config/status - Check status
- GET /api/jira/projects - Fetch all projects
- GET /api/jira/projects/:projectKey/user-stories - Fetch user stories
- GET /api/jira/issues/:issueKey - Fetch specific issue

Updated Files:

4. server.ts
   - Registered Jira router
   - Auto-loads Jira config from environment on startup
5. .env
   - Added Jira configuration variables:
     - JIRA_HOST
     - JIRA_EMAIL
     - JIRA_API_TOKEN
6. schemas.ts
   - Added Zod validation schemas for Jira data
   - Types: JiraConfig , JiraProject , JiraUserStory

Phase 2: Frontend Implementation ✅

Created Files:

Todos (12/12)

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JIRA INTEGRATION IMPLEMENTATION PLAN

Types: JiraConfig , JiraProject , JiraUserStory

Phase 2: Frontend Implementation ✅

Created Files:

7. JiraConfig.tsx
   - Form component for Jira credential configuration
   - Features:
     - Input fields for Host, Email, API Token
     - Connection validation
     - Status badge (Configured/Not Configured)
     - Help text with links to Jira token generation
     - Success/error messages
8. ProjectDropdown.tsx
   - Dropdown for selecting Jira projects
   - Auto-fetches and populates projects from Jira
   - Real-time loading states
   - Error handling
   - Triggered on Jira config success
9. UserStoryDropdown.tsx
   - Dropdown for selecting user stories
   - Auto-fetches stories when project is selected

Todos (12/12)

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- Input fields for Host, Email, API Token
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- Status badge (Configured/Not Configured)
- Help text with links to Jira token generation
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8. ProjectDropdown.tsx
   - Dropdown for selecting Jira projects
   - Auto-fetches and populates projects from Jira
   - Real-time loading states
   - Error handling
   - Triggered on Jira config success
9. UserStoryDropdown.tsx
   - Dropdown for selecting user stories
   - Auto-fetches stories when project is selected
   - Displays story metadata (status, description)
   - Cascading dropdown (depends on project selection)
   - Status badges with color coding

Updated Files:

10. api.ts
    - Added 5 Jira API methods:
      - configureJira()

Todos (12/12)

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JIRA INTEGRATION IMPLEMENTATION PLAN

Updated Files:

10. api.ts
    - Added 5 Jira API methods:
      - configureJira()
      - getJiraConfigStatus()
      - fetchJiraProjects()
      - fetchJiraUserStories()
      - fetchJiraUserStory()
11. types.ts
    - Added Jira types:
      - JiraProject
      - JiraUserStory
      - JiraConfigRequest
      - JiraConfigStatus
12. App.tsx
    - **Integrated all Jira components:**
      - Jira Config section at top
      - Project dropdown (auto-populated)
      - User Story dropdown (cascading)
      - Auto-population of form fields from selected story
      - Smooth user flow: Configure → Select Project → Select Story → Generate Tests

Todos (12/12)

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JIRA INTEGRATION IMPLEMENTATION PLAN

- Shows status and description on hover
- Automatically populates form fields when selected

✅ Critical Features:

- **Backend-First Architecture** - All Jira logic in backend, frontend consumes APIs
- **Error Handling** - Comprehensive error messages for debugging
- **Validation** - Zod schemas for request/response validation
- **Type Safety** - Full TypeScript typing throughout
- **Responsive UI** - Mobile-friendly components
- **Loading States** - Visual feedback during API calls
- **Security** - Basic Auth with API tokens (no credentials in logs)

🚀 How to Use

Setup:

1. **Add Jira credentials to** .env :

```
JIRA_HOST=https://your-domain.atlassian.net
JIRA_EMAIL=your-email@example.com
JIRA_API_TOKEN=your-api-token-here
```

Todos (12/12)

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- **Security** - Basic Auth with API tokens (no credentials in logs)

🚀 How to Use

Setup:

1. **Add Jira credentials to** .env :

```
JIRA_HOST=https://your-domain.atlassian.net
JIRA_EMAIL=your-email@example.com
JIRA_API_TOKEN=your-api-token-here
```

2. **Or configure via UI:**
   - Click "Configure Jira" button
   - Enter credentials
   - Click "Connect to Jira"

Workflow:

1. Configure Jira credentials (one-time setup)
2. Select project from dropdown (auto-populated)
3. Select user story from dropdown (auto-populated)
4. Form fields auto-populate from the story
5. Generate test cases as usual

Todos (12/12)

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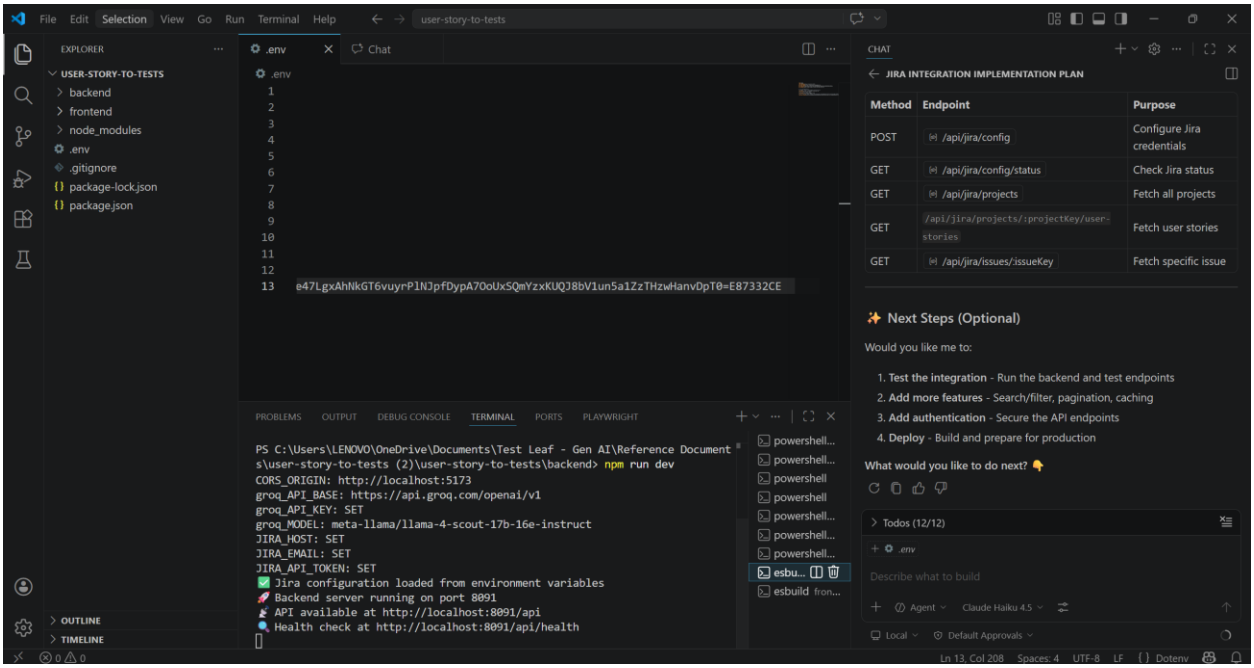
2. Select project from dropdown (auto-populated)
3. Select user story from dropdown (auto-populated)
4. Form fields auto-populate from the story
5. Generate test cases as usual

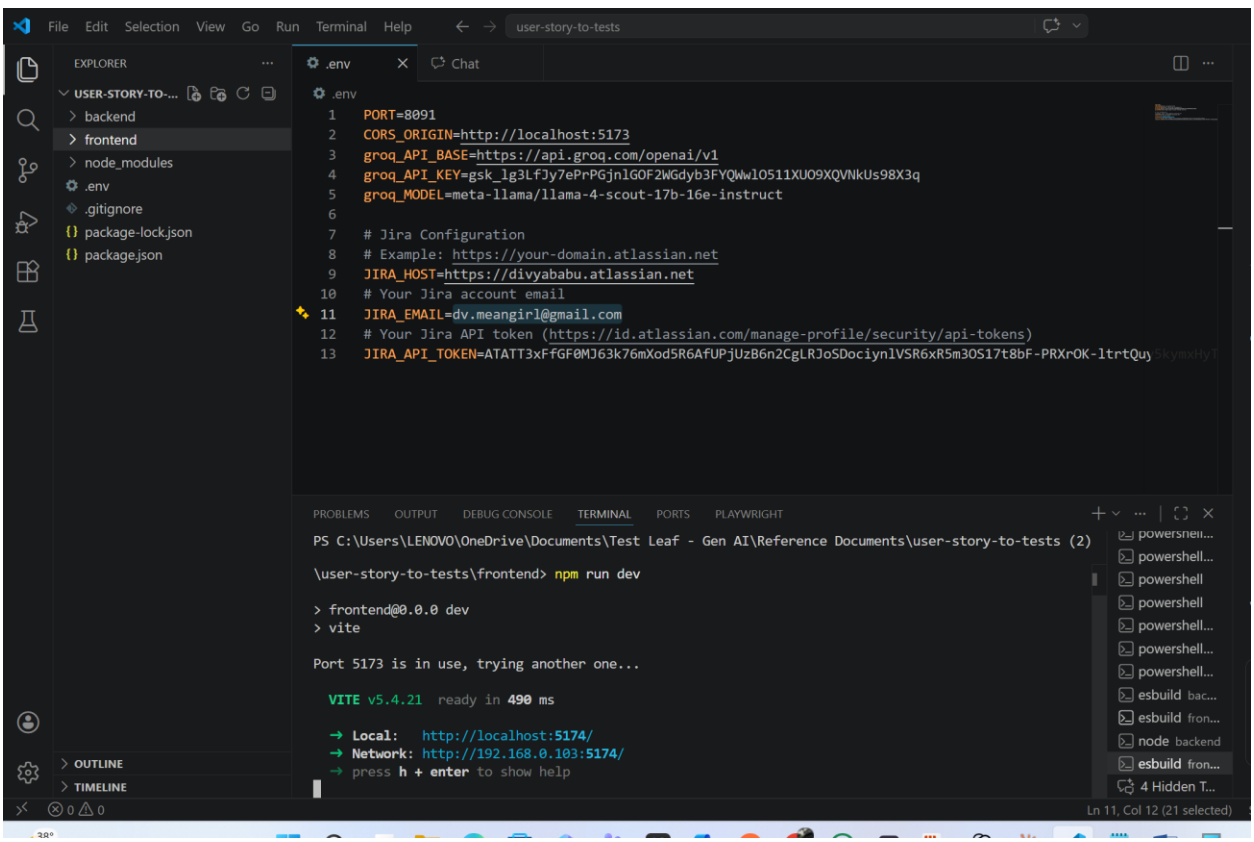
📊 API Endpoints Ready

Method	Endpoint	Purpose
POST	/api/jira/config	Configure Jira credentials
GET	/api/jira/config/status	Check Jira status
GET	/api/jira/projects	Fetch all projects
GET	/api/jira/projects/:projectKey/user-stories	Fetch user stories
GET	/api/jira/issues/issueKey	Fetch specific issue

Backend run successfully

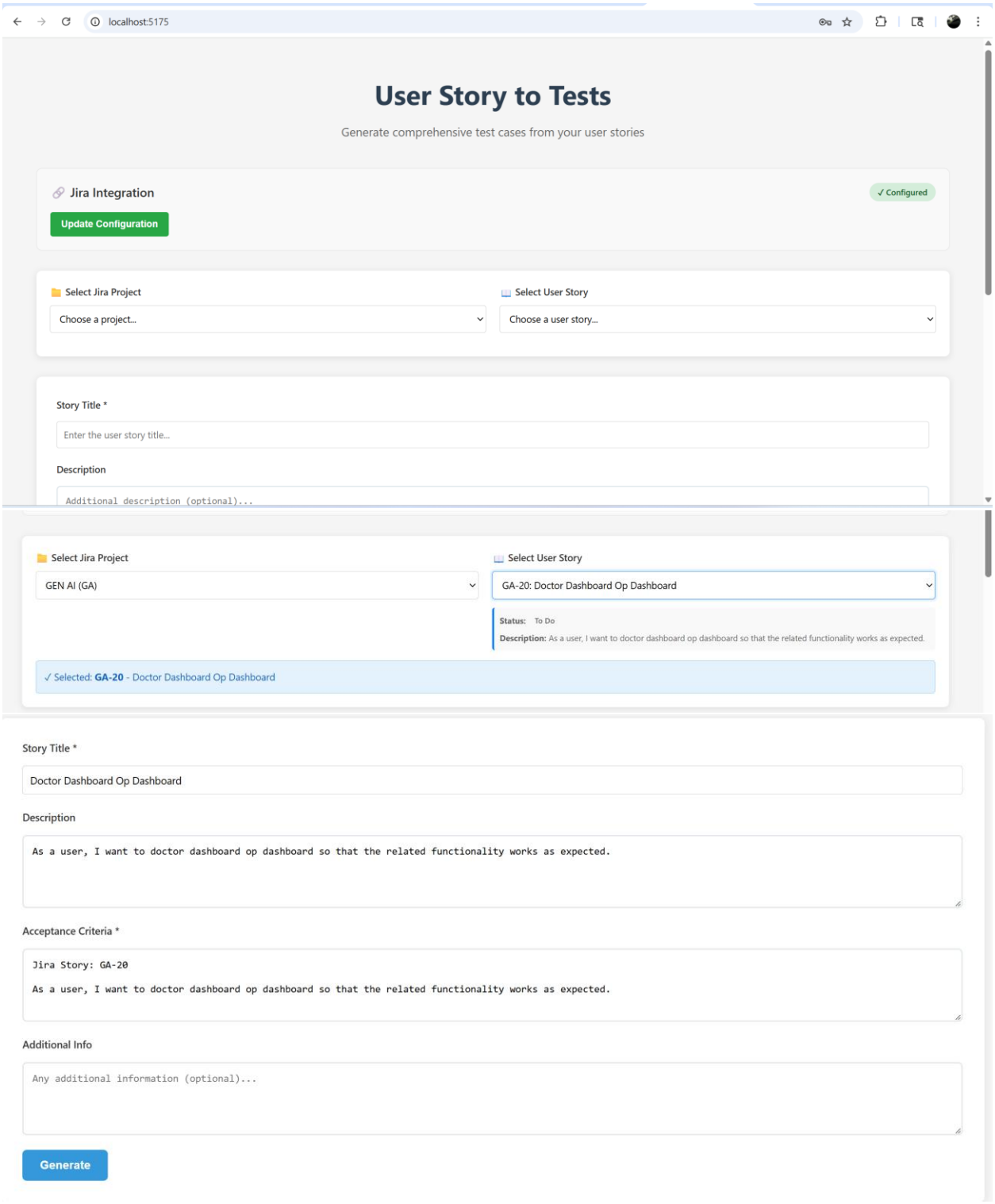


Front end also ran successfully



Had to debug and AI fixed few issues and the port was set to 5175

JIRA Integration tool



Generated Test Cases

20 test case(s) generated • Model: meta-llama/llama-4-scout-17b-16e-instruct • Tokens: 2797

Test Case ID	Title	Category	Priority	Expected Result
▸ TC-001	Doctor Dashboard OP Dashboard Access	Positive	High	OP Dashboard is displayed with no errors
▸ TC-002	Doctor Dashboard OP Dashboard Patient Search	Positive	High	Patient list is displayed with matching results
▸ TC-003	Doctor Dashboard OP Dashboard Patient Details	Positive	High	Patient details are displayed with no errors
▸ TC-004	Doctor Dashboard OP Dashboard No Patient Found	Edge	Medium	No patient found message is displayed
▸ TC-005	Doctor Dashboard OP Dashboard Invalid Credentials	Negative	High	Access is denied with an error message
▸ TC-006	Doctor Dashboard OP Dashboard Session Timeout	Edge	Medium	Session timeout error is displayed
▸ TC-007	Doctor Dashboard OP Dashboard Multiple Patient Selection	Positive	Medium	Patient details are displayed for each selected patient
▸ TC-008	Doctor Dashboard OP Dashboard Patient Filter	Positive	Medium	Patient list is updated with filtered results
▸ TC-009	Doctor Dashboard OP Dashboard Export Patient List	Positive	Low	Patient list is exported successfully
▸ TC-010	Doctor Dashboard OP Dashboard Print Patient Details	Positive	Low	Patient details are printed successfully
▸ TC-011	Doctor Dashboard OP Dashboard Patient Details Edit	Positive	High	Patient details are updated successfully
▸ TC-012	Doctor Dashboard OP Dashboard Patient Delete	Positive	High	Patient is deleted successfully
▸ TC-013	Doctor Dashboard OP Dashboard Access Denied	Negative	High	Access is denied with an error message
▸ TC-014	Doctor Dashboard OP Dashboard SQL Injection	Negative	High	Error is displayed and no data is compromised
▸ TC-015	Doctor Dashboard OP Dashboard Cross-Site Scripting (XSS)	Negative	High	Error is displayed and no data is compromised
▸ TC-016	Doctor Dashboard OP Dashboard Patient Search with Special Characters	Edge	Medium	Patient list is displayed with matching results
▸ TC-017	Doctor Dashboard OP Dashboard Patient Search with Numbers	Edge	Medium	Patient list is displayed with matching results
▸ TC-018	Doctor Dashboard OP Dashboard No Internet Connection	Edge	Medium	Error is displayed

TC-013	Doctor Dashboard OP Dashboard Access Denied	Negative	High	Access is denied with an error message
TC-014	Doctor Dashboard OP Dashboard SQL Injection	Negative	High	Error is displayed and no data is compromised
TC-015	Doctor Dashboard OP Dashboard Cross-Site Scripting (XSS)	Negative	High	Error is displayed and no data is compromised
TC-016	Doctor Dashboard OP Dashboard Patient Search with Special Characters	Edge	Medium	Patient list is displayed with matching results
TC-017	Doctor Dashboard OP Dashboard Patient Search with Numbers	Edge	Medium	Patient list is displayed with matching results
TC-018	Doctor Dashboard OP Dashboard No Internet Connection	Edge	Medium	Error is displayed
TC-019	Doctor Dashboard OP Dashboard Browser Compatibility	Edge	Low	Error is displayed
TC-020	Doctor Dashboard OP Dashboard Page Load Time	Edge	Low	Page load time is within acceptable limits

Export Test Cases

JSON CSV Excel